SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

COMPUTER SCIENCE AND ENGINEERING PROGRAMME

CSA3717 – SOFTWARE TESTING

EXPERIMENT 1

SYSTEM SPECIFICATION REVIEW AND SOFTWARE DEFECT IDENTIFICATION – ATM SYSTEM

1. Aim:

An Automated Teller Machine (ATM) allows bank customers to perform cash withdrawal, balance enquiry, fund transfer, PIN change and mini-statement operations using a valid bank card and PIN. The ATM communicates with the bank server to authenticate the customer and authorize transactions.

2. Main System Specifications

1. Only active bank cards should be accepted.
2. Customers must enter a valid PIN before accessing account services.
3. Cash withdrawal must be allowed only when the account has sufficient balance and the ATM has sufficient cash.
4. Withdrawal amount must satisfy the bank's minimum, maximum and denomination rules.
5. Balance enquiry must display the latest available balance.
6. Every successful transaction must update the bank ledger and generate a receipt when requested.
7. After repeated incorrect PIN attempts, the card or session must be blocked according to bank policy.
8. The ATM must return the card and end the session after inactivity or customer logout.

Problem Statement:

An Automated Teller Machine (ATM) system allows customers to perform banking operations such as cash withdrawal, balance enquiry, fund transfer, PIN change, and mini-statement generation. However, software defects may occur in authentication, transaction processing, cash dispensing, balance updating, and session management. The objective is to examine the ATM system specifications, identify possible software defects, classify them based on their severity and type, and suggest suitable solutions to improve the reliability, security, and performance of the system.

3. Identified / Potential Software Defects

The following defects are representative issues that may be discovered during specification-based testing. They are examples for the case study and should be confirmed through execution in a real ATM test environment.

Defect ID	Module	Observed / Potential Defect	Severity	Impact	Suggested Correction
ATM-01	Authentication	ATM accepts an incorrect PIN after the third failed attempt.	Critical	Unauthorized account access may occur.	Block the session/card after the configured retry limit.
ATM-02	Cash withdrawal	ATM debits the account but does not dispense cash.	Critical	Customer loses access to funds and reconciliation is required.	Use transaction reversal and cash-dispense confirmation.
ATM-03	Balance update	Displayed balance is not refreshed after a withdrawal.	High	Customer receives incorrect account information.	Fetch the updated balance after transaction commit.
ATM-04	Amount validation	ATM accepts an amount that is not a valid note denomination.	Medium	Cash dispensing may fail or produce an incorrect amount.	Validate amount against supported denominations.
ATM-05	Session timeout	Session remains active after prolonged inactivity.	High	Another person may access the account.	Terminate inactive sessions and return the card.
ATM-06	Receipt	Receipt prints an incorrect transaction date/time.	Medium	Customer records become unreliable.	Synchronize ATM clock with trusted server time.
ATM-07	Cash availability	ATM permits withdrawal when cassette cash is insufficient.	High	Transaction failure and customer inconvenience.	Check dispenser cash availability before authorization.
ATM-08	Fund transfer	Transfer is shown as successful although beneficiary account is invalid.	Critical	Funds may be misdirected or transaction status may be inconsistent.	Validate beneficiary and confirm ledger posting atomically.

4. Sample Defect Report

Field	Description
Bug ID	ATM-02
Title	Account is debited but ATM does not dispense cash.
Precondition	Customer has sufficient balance; ATM dispenser is configured for testing.
Steps	1. Insert a valid card. 2. Enter correct PIN. 3. Select cash withdrawal. 4. Enter a valid amount. 5. Simulate dispenser failure after debit.
Expected Result	If cash is not dispensed, the transaction should be reversed and the account balance restored.
Actual Result	The account is debited, but no cash is dispensed and no automatic reversal occurs.
Severity / Priority	Critical / High
Environment	ATM simulator, bank test server, test card and cash-dispenser simulator.
Suggested Fix	Use a transaction state mechanism that confirms cash dispensing and reverses the debit when dispensing fails.
Status	Open – illustrative sample.

5. Defect Classification

Classification	Meaning	ATM Example
Functional	A feature does not meet its requirement.	Balance enquiry shows the wrong balance.
Security	Authentication or authorization is weak.	Incorrect PIN is accepted.
Performance	Response time is unacceptable.	Transaction takes too long to complete.
Usability	The interface is confusing or difficult to use.	Error message does not explain the problem.
Interface / Integration	Communication between ATM and bank server fails.	ATM shows success when server posting fails.
Data / Transaction	Stored or updated data is incorrect.	Account is debited without cash dispensing.

6. Testing Approach

- Review the ATM specification and identify measurable expected behaviour.
- Prepare valid, invalid, boundary and recovery test data.
- Execute tests using an ATM simulator and a controlled bank test environment.
- Record test evidence, transaction IDs, timestamps and screenshots where permitted.

- Prioritize defects based on customer impact, security risk and financial impact.
- Retest fixes and perform regression testing on related transaction functions.

7. Conclusion

Specification-based testing of an ATM focuses on transaction correctness, security, cash handling, session control and reliable communication with the bank server. Early identification and classification of defects reduces financial risk and improves customer trust.